

Autonomous Battery Units as an Enabling Technology for Urban Air Mobility

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Urban air mobility (UAM), which transports a few passengers or a small amount of cargo within a metropolitan area on demand, expects future aircraft — e.g., electric vertical takeoff and landing (eVTOL) aircraft — to use batteries for energy during flight. However, battery energy density remains many times smaller than the threshold required for practical and economical flight, and commercially-viable battery energy density has not increased significantly over the last decade. To hasten electrification of flight, we propose autonomous battery units (ABUs) as an enabling technology for UAM with eVTOLs. An ABU consists of one or more rotors, a battery, a computer system with sensors for autonomous flight, and an attitude control system inside a chassis that detaches from and attaches to an aircraft before, during, and after flight. In this work, we quantify the benefits of ABUs in eVTOL UAM operations, motivating subsequent design and autonomy work. ABUs provide assisted takeoffs, extended flight time, safer landings, and more economical flight for the short-duration, common case of UAM. Assisted-takeoff ABUs reduce mission energy by seven to 24%; assisted-cruise ABUs reduce end-of-mission depth-of-discharge (DOD) to as little as 16%; assisted-landing ABUs support divert missions with as little as 30 kWh mission energy capacity; and ABUs increase UAM flights per eight-hour period by 1.5 to 2.6 $\times$, showing that ABUs significantly improve eVTOL operations for UAM.

I. Introduction

Emerging, electric vertical takeoff and landing (eVTOL) aircraft support new urban air mobility (UAM) concepts for transporting people and cargo across metropolitan areas. UAM aims to alleviate urban congestion, reduce travel time, and create new modes of air transportation. Many UAM applications aim to avoid street traffic congestion while delivering cargo or transporting people, e.g. between high-rises and airports. Such services operated in the United States — Los Angeles and New York City in particular — for over two decades using gas-powered helicopters until the 1970s; both services ended after repeated safety incidents [1]. Charter VTOL transportation services still operate in New York City and other urban areas throughout the world. However, economics, regulatory factors, and other community limitations often restrict flight frequency. Electric VTOL aircraft aim to address UAM challenges (e.g., fuel availability and noise), but technical limitations (e.g., battery energy density) prevent widespread adoption.

For example, eVTOL aircraft carry more weight compared to gasoline-powered counterparts due to dense, heavy batteries. Previous work shows that eVTOL takeoff gross weight depends strongly on battery energy density [2]. At the same time, eVTOL aircraft store less energy for flight; the energy density by mass of commercial batteries is 50 $\times$ less than aviation gasoline. While some research efforts suggest the possibility of better battery energy density in the future, commercially-viable batteries exhibit little improvement in energy density over the past decade (see Section II.C). Other electrified aircraft proposals face similar challenges regarding battery weight and energy capacity. For example, regional air mobility (RAM) aircraft cannot operate from batteries without a 4 $\times$ to 5 $\times$ increase in energy density by mass, and even this improvement provides only one-tenth the range of a comparable, conventional aircraft [3]. Charging capacity at eVTOL vertiports also limit UAM operations [4]. Despite clear demand and applications for eVTOL UAM, technical support in the form of sufficient battery energy density might not arrive for decades.

We circumvent this limitation through use of a different technical advancement — namely, autonomy — in conjunction with batteries to make eVTOL UAM feasible in the near term. We introduce the concept of an *autonomous battery unit*, or ABU, that attaches to and detaches from an aircraft during flight and flies independently from the host

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Fig. 1 UAM vehicles. Left: helicopters (credit); Sikorsky S-76 (Don Ramey Logan), Bell 407 (Raimond Spekking), Airbus H130 (Zidane Hartono). All images CC BY-SA 4.0. Right: electric aircraft (credit); Joby Aviation aircraft (public domain); Archer Midnight (Archer Aviation); Supernal S-A2 (Supernal).

vehicle. An ABU consists of one or more rotors, a battery, a computer system with sensors for autonomous flight, and an attitude control system inside a chassis. ABUs improve electric flight during takeoff, cruise, and landing. During takeoff, which is one of the most power-intensive portions of a flight profile, ABUs provide energy so that batteries built into the electric aircraft need not be depleted. After completing takeoff, these ABUs detach and autonomously return to the departure site to recharge. During flight, depleted ABUs detach from the electric aircraft and autonomously land at intermediate sites to recharge. Similarly, charged ABUs takeoff from intermediate sites and autonomously rendezvous with aircraft to act as a form of electric, in-flight refueling. In the event of a failure during landing, ABUs autonomously rendezvous with aircraft to support unplanned divert missions. Because eVTOLs must be designed to support a maximum flight duration, but rarely fly missions at this maximum distance, ABUs allow for smaller starting masses in the common mission case, increasing performance and economic viability. We expect distributed electric propulsion (DEP) literature [5–11] to strongly overlap with ABU applications.

Results show that ABUs (i) alleviate the power burden of takeoff, reducing total mission energy by seven to 24%; (ii) alleviate the energy burden of cruise through rendezvous with charged batteries and detachment of depleted batteries, reducing onboard battery end-of-mission depth-of-discharge (DOD) to as little as 16%; (iii) improve landing safety by supporting unplanned divert missions with as little as 30 kWh mission energy capacity; and (iv) better support the common case of short-duration UAM flights by increasing flights per eight-hour period by 1.5 – 2.6×. To evaluate ABU performance, this work develops and utilizes a novel, *evtolpy* simulation and sizing software framework released as an open-source artifact at the following URL: <https://github.com/starbelt/evtolpy/releases/tag/scitech26>

II. Background

A. Urban air mobility

Urban air mobility (UAM) transports passengers or cargo within a metropolitan region on demand. Passenger count ranges from three to 30, and flights last less than 60 minutes. For example, a UAM service operated multiple Sikorsky S-61L helicopters within an 80 km radius of New York City during the 1970s [12], each seating up to 25 passengers. Today, a UAM service operates Bell 407 (five passengers), Sikorsky S-76 (up to 12 passengers), and Airbus helicopters (four to six passengers), and other commuter aircraft [13]. Recent eVTOL aircraft include the Archer Midnight [14] and Joby Aviation [15] aircraft. Figure 1 illustrates aircraft commonly deployed for UAM.

In urban environments, limited space for takeoff and landing encourages use of VTOL aircraft like helicopters for UAM. Figure 1, top left, illustrates helicopters commonly deployed for UAM. However, helicopters generally operate with a single, primary engine, making failures catastrophic. Many UAM services cease operations after repeated safety incidents involving helicopters [1]. Additionally, helicopters fly inefficiently during cruise. Despite the benefits of VTOL, inefficient cruise flight makes helicopters economically impractical for longer trips.

Emerging eVTOL aircraft (e.g., Archer Midnight [14], Joby S4 [15]) combine VTOL capabilities of a helicopter with efficient cruise capabilities of a winged aircraft. Figure 1, right, illustrates these aircraft. Electric VTOL aircraft access urban environments with limited takeoff and landing area, and they cruise farther using winged designs. Figure 2 illustrates an eVTOL UAM mission profile [16]. These missions begin with taxi, hover climb, transition climb, and depart procedures. The eVTOL accelerates, climbs, cruises, and decelerates to descend. Finally, the eVTOL performs arrival procedures, transition descend, hover descend, and arrival taxi. Each mission profile includes reserve segments.

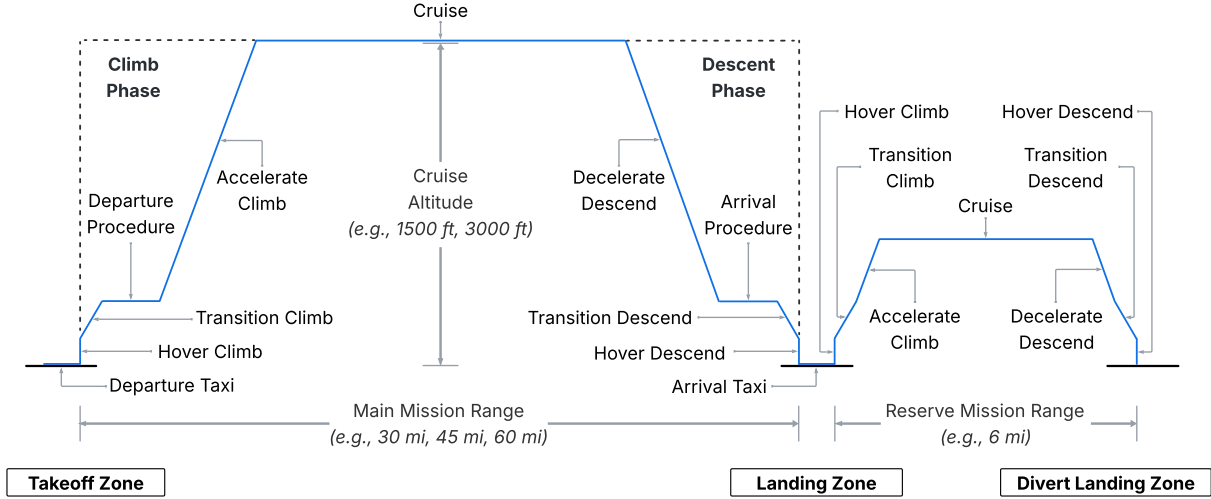
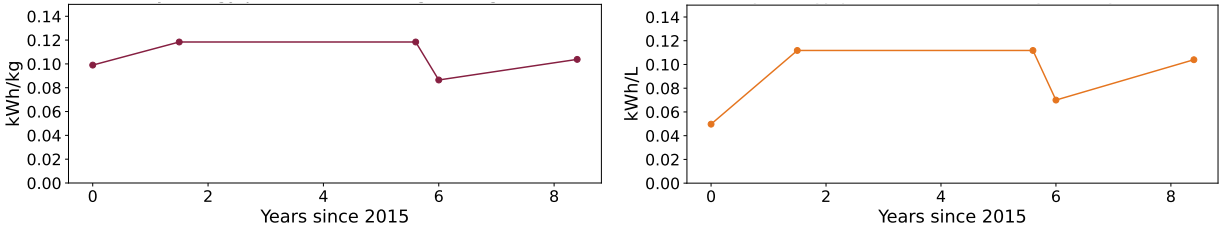


Fig. 2 This work considers three flight ranges (30, 45, and 60 miles) and two cruise altitudes (1500 and 3000 ft).



(a) A decade of energy per unit mass for commercial batteries **(b)** Energy per unit volume during the same time period

Fig. 3 Commercially-viable batteries show little energy density change when accounting for integration factors.

B. Sizing eVTOL aircraft

Aircraft development generally begins with conceptual sizing frameworks that balance mission requirements, vehicle performance, and system-level constraints. For conventional fixed-wing aircraft, methods such as Raymer [17] provide rapid estimates of takeoff gross weight (TOGW), propulsion needs, and fuel burn by dividing the mission into segments (takeoff, climb, cruise, loiter, reserve) and iterating on TOGW until meeting fuel requirements. These methods tailor to fuel-based, fixed-wing aircraft and do not translate directly to eVTOL vehicles.

Key differences in eVTOL physics break core assumptions of traditional sizing. Fuel mass decreases throughout flight, but batteries maintain constant mass regardless of energy expended, fundamentally altering the weight-energy coupling in the sizing loop. Vertical operations introduce hover and transition phases characterized by short-duration, high-power demand that fixed-wing tools do not represent well [18]. Integration of the battery system also requires new considerations, including structural packaging and energy reserves, that are absent from existing frameworks. Because batteries have far lower energy density than fuel [19, 20], energy budgeting, not range or payload fraction, becomes the dominant constraint in eVTOL design. As a result, vehicle feasibility is highly sensitive to battery characteristics [21], which substantially influences both aircraft weight and achievable mission performance [22, 23]. Recent work introduces eVTOL-specific sizing [2, 24–34]. Yet, the field still lacks a simple and fast TOGW estimation method analogous to the widely-used Raymer framework for fixed-wing aircraft. This work fills that gap by developing *evtolpy*, a simulation and sizing framework that blends physics-based methods with empirical weight regressions (see Section III).

C. Battery technology for electric aircraft

Battery energy density remains several times smaller than the threshold required for practical and economical flight. While some research efforts suggest the possibility of sufficient battery energy density in the future, commercially-viable batteries exhibit little improvement in energy density over the past decade. Aviation gasoline, or avgas, holds an energy

mass density of about 12 kWh/kg; for example, the commonly-used 100LL avgas contains 12.14 kWh/kg. With a density of 0.72 kg/L at 15 °C and 1 atm, the corresponding energy volume density is 8.74 kWh/L. In comparison, commercially-viable batteries contain an energy mass density of 0.244 kWh/kg and an energy volume density of 0.163 kWh/L. Thus, batteries are 50× less energy-dense in mass and 54× less energy-dense in volume than aviation gasoline. These numbers are optimistic; after accounting for operational inefficiencies due to power conversion, aging, and additional components to deliver power, the energy density gap increases.

Figure 3 illustrates energy density by mass and volume of commercially-available, wall-mounted battery storage over the past decade. These products more clearly illustrate the mass and volume overhead of additional hardware required to deliver useful, A/C power from commercial batteries. These battery units vary between 120 L and 200 L in volume and 10 kW h to 13.5 kW h in energy capacity. Across a decade of products, the energy density relative to both mass and volume shows no significant change. The maximum effective energy density by mass is less than 0.12 kWh/kg, and the maximum effective energy density by volume is also less than 0.12 kWh/L.

This large performance gap gives rise to active research quantifying, evaluating, and addressing the energy density challenge. Accurate simulation and prediction of battery energy storage and delivery remain open and active areas of research [35–38]. Simulations of emerging, electric powertrains and energy-efficient trajectories find means for reducing energy consumption during flight [39]. These proposals compare against baseline models that represent non-electric aircraft expected to enter service during the next decade [40]. Others attempt to carefully balance static and dynamic battery characteristics to maximize potential performance [41]. Because of the large gap in energy density between aviation gasoline and batteries, some investigate the application of “sustainable aviation fuel” (SAF) as an alternative to both options [42]. Others evaluate hybrid energy approaches, which combine use of avgas, electric batteries, and fuel cells [43–48]. Many concepts consider hydrogen fuel cells to augment energy storage during flight [49–51]. Some concepts investigate charging batteries during flight, either using solar arrays [7, 52] or regenerative charging during cruise or descent [53]. We propose autonomous battery units that attach to and detach from aircraft during flight. Instead of depending on future improvements to battery energy density, ABUs leverage currently-available autonomy technology to rendezvous with electric aircraft and replenish depleted, electric energy stores during flight.

III. The `evtolpy` Simulation and Sizing Software Framework: Motivating the Need for ABUs

This work develops `evtolpy`, a simulation and sizing software framework for eVTOLs, which blends physics-based methods with empirical mass regressions. Sharing a software design with analogous simulation frameworks for autonomous space systems [54], and building on previous review work [55], `evtolpy` computes power and energy requirements for each mission segment as functions of mass, aerodynamics, and propulsion, while an iterative loop updates vehicle component weights. Users vary mission parameters (e.g., range, loiter time, payload), aircraft characteristics (e.g., rotor count, wing aspect ratio), and type (e.g., tilt-wing, lift-plus-cruise, multi-copter) to quickly assess impacts on vehicle feasibility and TOGW.

The `evtolpy` framework models each mission as a sequence of segments and computes shaft and electric power for each segment based on aerodynamic forces, rotor performance, and vehicle mass. Velocity and duration inputs determine segment kinematics, and drag forces are computed from induced and parasitic components, including contributions from the fuselage, wing, empennage, landing gear, and (when applicable) stopped-rotors. Parasitic drag is computed by aggregating the zero-lift drag contributions of each component as follows:

$$C_{D0} = C_{D0, \text{fuselage}} + C_{D0, \text{wing}} + C_{D0, \text{empennage}} + C_{D0, \text{landing gear}} + C_{D0, \text{stopped-rotor}} \quad (1)$$

Shaft and electric power depend on the horizontal and vertical force components F_{horiz} and F_{vert} , the associated velocities v_{horiz} and v_{vert} , the rotor efficiency η_{rotor} , and the electric propulsion unit (EPU) efficiency η_{epu} . These quantities define average shaft power, the resulting average electric power, and the energy consumption of a single mission segment E_{segment} , which is later used to generate the entire mission profile shown in Figure 2.

$$P_{\text{shaft, avg, i}} = \frac{F_{\text{horiz}} v_{\text{horiz}} + F_{\text{vert}} v_{\text{vert}}}{\eta_{\text{rotor}}} \quad P_{\text{elec, avg, i}} = \frac{P_{\text{shaft, avg}}}{\eta_{\text{epu}}}$$

As an example, consider the Transition Descend (TD) segment, which contains vertical and horizontal motion and thus illustrates each shaft-power calculation component. Here, average shaft power accounts for hover-induced assist

power needed to balance net vertical force and aerodynamic power for horizontal flight. This yields

$$P_{\text{shaft,avg,TD}} = \frac{P_{\text{hover}} + F_h v_h}{\eta_{\text{rotor}} W_{\text{kW}}} = \frac{\left[(W - L + m a_v) \sqrt{\frac{W - L + m a_v}{2 \rho A_{\text{disk}}}} \right] + (D_{\text{total}} + m a_h) v_h}{\eta_{\text{rotor}} W_{\text{kW}}}, \quad (2)$$

where W is the aircraft weight, L is the aerodynamic lift, m is the vehicle mass, a_v and a_h are the prescribed vertical and horizontal accelerations, ρ is the ambient air density, A_{disk} is the total rotor disk area, D_{total} is the total drag, v_h is the horizontal velocity, η_{rotor} is the rotor efficiency, and W_{kW} is the watts-to-kilowatts conversion factor.

Equivalent formulas for all remaining mission segments — each reflecting the specific kinematics and aerodynamic characteristics of that segment — are documented in the online `evtolpy` reference available at the following URL: <https://github.com/starbelt/evtolpy/releases/tag/scitech26>.

Integrating electric power over each mission segment yields its segment-level energy consumption, denoted E_i , where the index i refers to the i th segment of the mission profile. These segment energies form the building blocks of the total mission energy, as each segment has its own average power demand and duration. Summing the segment energies across all main-mission and reserve-mission segments gives the total required battery energy E_{mission} . This mission-based sizing model produces the segment-level power and energy distributions used throughout the study, including those illustrated in Figure 4.

$$E_i = P_{\text{elec,avg},i} \cdot t_i \quad E_{\text{mission}} = \sum_{i=1}^N E_i$$

These segment-level loads are then input to an iterative aircraft sizing loop that updates structural, subsystem, and battery masses until the maximum takeoff weight (MTOW) converges. Component weights — wings, rotors, hubs, fuselage, landing gear, empennage, and boom structural — are computed using empirical regressions adapted from the NASA NDARC rotorcraft design database [56]. Additional subsystem masses, including the electric-propulsion unit, actuators, avionics, environmental-control system, furnishings, and high- and low-voltage power-distribution systems, are incorporated into the empty-weight calculation along with a fixed mass margin. Because many of these component weights scale with MTOW through the NDARC regressions or through internal sizing relationships, the empty weight itself becomes a function of MTOW. Likewise, the required battery mass depends on MTOW through its influence on power and energy demand. These coupled dependencies motivate the iterative sizing procedure used in the simulator. After evaluating the required mission energy, `evtolpy` computes the corresponding battery mass using:

$$m_{\text{battery}} = \frac{E_{\text{mission}}}{e_{\text{spec}}}. \quad (3)$$

where e_{spec} is battery specific energy. Accessible energy fraction margins and pack-integration factors are applied to account for practical limits on usable capacity. The resulting battery mass is incorporated into the MTOW update:

$$W_{\text{MTOW,new}} = W_{\text{empty}}(\text{MTOW}) + W_{\text{payload}} + W_{\text{battery}}(\text{MTOW}). \quad (4)$$

Eq. 4 is implicit and requires an iterative solution converging on MTOW. In each iteration, MTOW is used to compute the new battery weight and the empty weight. The convergence criterion is set as:

$$|W_{\text{MTOW,new}} - W_{\text{MTOW,prev}}| < \epsilon, \quad \epsilon = 10^{-3}. \quad (5)$$

The `evtolpy` simulator accepts inputs describing the aircraft geometry (e.g., wingspan, fuselage dimensions, rotor diameter), aerodynamic and drag-related properties, propulsion system characteristics (e.g., rotor count, thrust and power coefficients, efficiency parameters), mission definitions (e.g., segment speeds, durations, and vertical profiles), battery cell and pack properties (e.g., specific energy, accessible energy fraction, pack integration factor), and relevant environmental conditions such as air density and atmospheric viscosity. A summary list of important input and output parameters of the simulator is provided in Table 2.

We validate this framework against published data for representative eVTOLs, including the Archer Midnight, Joby S4, and Supernal S-A2 [57], and find close agreement in MTOW for comparable mission definitions. For Archer Midnight in particular, the `evtolpy` baseline model is sized to match the advertised MTOW, payload, cruise speed, and nominal cruise altitude, and produces a practical mission range of approximately 90 mi when mission procedures and

Table 1 Comparison of Archer Midnight reference data [58] and evtolpy baseline model.

Quantity	Archer Midnight (public data)	evtolpy model
MTOW [kg]	3175	3175.2
Payload [kg]	453	453
Cruise speed [kt]	130	130
Nominal cruise altitude [ft]	2000	2000
Maximum cruise range [mi]	~100	~89.2

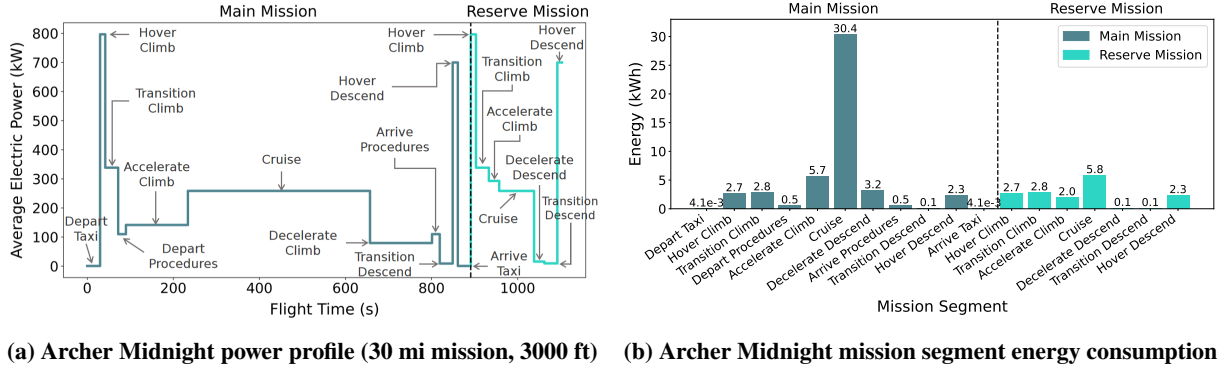


Fig. 4 Left: Power for flight peaks during hover segments. Right: Energy for flight peaks during cruise.

reserves are included (see Table 1). This value closely matches the 100 mi range reported by the manufacturer and is consistent with expected differences between conceptual design assumptions and marketing specifications.

Because evtolpy provides a unified and internally-consistent mapping from mission requirements to aircraft mass, power, and energy, it is well-suited for evaluating architectural changes such as ABUs, whose benefits depend on segment-specific loads and battery sizing behavior.

To motivate the need for ABUs in achieving practical near-term electric flight, this work analyzes power and energy demands of each mission segment for representative eVTOL aircraft. We model three vehicles: the Archer Aviation Midnight, Joby Aviation S-4, and Supernal S-A2. For each aircraft, this work evaluates flights of 30 mi, 45 mi, and 60 mi and assesses two cruise altitudes: 1500 ft and 3000 ft.

Figures 4a and 4b illustrate results for a single case: the Archer Midnight flying a 30 mi mission at a 3000 ft cruise altitude. The power profile shows that the highest instantaneous power draws occur during hover, climb, and descent, when the rotors generate the full weight-supporting lift. The energy profile reveals that cruise dominates total mission energy, accounting for nearly half of total energy when including reserves. Without reserve, cruise represents the vast majority of mission energy due to its long duration relative to other segments.

These numbers highlight a fundamental challenge for eVTOL operations. A monolithic onboard battery must simultaneously satisfy the peak power needed for vertical flight and the large energy required for cruise. Larger batteries increase aircraft mass, which in turn raises hover power and further increases energy demand. Deep discharge levels extend charging time and reduce the number of flights achievable within an operating window. These limitations are especially problematic for high-frequency UAM operations. Even with continued improvements in battery and propulsion technology, near-term gains in energy density are insufficient to fully overcome these constraints.

This motivates the exploration of system-level alternatives that decouple mission energy from onboard battery mass. ABUs provide one such approach: by supplying supplemental energy during mission segments with high demand, ABUs reduce integrated battery size, limit depth of discharge, shorten charging time, and expand per-aircraft throughput, ultimately improving operational feasibility and economic viability.

IV. Autonomous Battery Units

ABUs solve energy-storage challenges for electric aircraft. As described in Section II.C, commercially-viable integrated battery systems provide, in the worse case, up to two orders of magnitude less energy per mass and volume than

Table 2 Summary of key eVTOLpy input and output parameters

Input Parameters	Output Parameters
Maximum Takeoff Weight (MTOW)	Updated Maximum Takeoff Weight (MTOW)
Payload Mass	Hover Thrust Coefficient (C_T)
Wingspan, Wing Taper Ratio, Aspect Ratio	Rotor Solidity
Fuselage Length, Width, Height	Rotor RPM (Hover)
Horizontal/Vertical Tail Volume Coefficients	Structural and Non-Structural Weight
Stall Speed	Wingspan and Wing Reference Area
Airfoil Drag Coeff. at Cruise C_L	Wing Loading
Maximum Lift Coefficient ($C_{L,max}$)	Wing Parasite Drag Coeff. (C_{D0})
Rotor Count, Lift/Tilt Rotor Allocation	Mean Aerodynamic Chord ($\bar{c}$)
Rotor Diameter, Tip Mach Number	Reynolds Number (Cruise and Stall)
Average Rotor Coefficient of Lift (C_L)	Wing Weight
Rotor Efficiency, Hover Power Efficiency	Fuselage Fineness Ratio
Trim Drag and Excrescence Factors	Fuselage Skin Friction Coeff. (C_f)
Landing Gear Drag Area	Fuselage C_{D0}/C_f Ratio
Stopped-Rotor Area Ratio	Fuselage Wetted/Reference Areas
Actuator, Avionics, ECS, Furnishings Masses	Fuselage Drag Area (C_DA)
High/Low Voltage Distribution Masses	Horizontal/Vertical Tail Weight and C_{D0}
Mass Margin Factor	Landing Gear C_{D0} and Weight
Battery Specific Energy	Boom Drag Area (C_DA) and Weight
Battery Accessible Energy Fraction	Total Rotor Disk Area
Battery Integration Factor	Disk Loading
Atmospheric Density, Viscosity Profiles	Lift Rotor and Hub Weight
Segment Speeds and Durations (All Mission Phases)	Tilt Rotor and Hub Weight
Reserve Mission Segment Definitions	Electric Propulsion Unit (EPU) Weight

aviation gasoline. Further, unlike gasoline, batteries do not decrease in mass when energy depletes. As a result, the mass of an electric aircraft averaged over a flight exceeds that of a comparable, gasoline-powered aircraft — requiring more energy for flight and exacerbating the energy-storage challenge. Thus, electric aircraft face two primary, energy-storage challenges: a dearth of stored, electric energy and excess mass in flight. ABUs address the lack of stored energy by replenishing energy buffers during flight; they address excess mass by detaching nearly-depleted batteries midflight.

These solutions echo analogous approaches for demanding flight profiles encountered by gasoline-powered military aircraft. These aircraft replenish energy stores with aerial refueling [59]. Drop tanks, or external auxiliary fuel tanks, also increase flight duration and, to reduce mass and drag after depletion, may be ejected. A historical lack of autonomy prevents widespread use of these solutions; for example, drop tanks generally are not recovered after ejection and instead crash into the ground. Some small-scale UAVs evaluate battery-swapping and docking concepts [60–65], showing feasibility of mid-mission energy replenishment, while ABUs target aircraft supporting UAM and longer flights. Maturation of autonomous capabilities, along with the much less volatile nature of batteries compared to gasoline, provide a new opportunity to address these challenges faced by emerging, electric aircraft.

A. ABU concept of operations

An autonomous battery unit (ABU) consists of one or more rotors driven by an electric motor, an attitude determination and control system (ADACS), and a battery array, all joined in a chassis. Design characteristics depend on the class of aircraft the ABU supports; this work quantifies the benefits of ABUs in eVTOL UAM operations, motivating subsequent design and autonomy work. The key, technical advancement making ABUs practical today is autonomy; vehicles now independently navigate the world in real time with precision. Examples of such navigation occur in

everyday life with self-driving electric vehicles (EVs) and rockets that return to and land at launch sites. ABUs use autonomous capabilities to shift electrified aviation from potentially more than a generation in the future to the present.

Constraints on stored energy capacity determine a key design concept principle: an ABU should expend minimal energy while delivering charged batteries to an in-flight aircraft. Similarly, an ABU should expend minimal energy to detach from an aircraft and land at a recharging station; most energy expenditure occurs while the ABU provides thrust or lift when attached to an aircraft. Another key design concept principle, minimizing mass, avoids adding excessive weight to an aircraft. These constraints inform a relatively barebones, direct-rendezvous design concept without wings for ABU lift. Such a direct-rendezvous ABU cannot efficiently travel significant lateral distances and thus supports (i) assisted liftoff before detaching and returning to the takeoff site; (ii) detachment during cruise and descending to a nearby landing site; (iii) direct ascent from an intermediate site near the flight path to attach a nearly-charged battery to an aircraft; and (iv) assisted landing in the event of a safety issue. In some cases (e.g., longer flights by a utility or RAM aircraft as opposed to a UAM aircraft), an ABU may need to travel greater lateral distances. For these scenarios, a blended-wing ABU design concept supports efficient travel across lateral distances. These key design principles for an ABU concept of operations (CONOPS) inform the evaluation and results in Section VI.

The onboard ADACS consists of sensors (e.g., cameras, lidar, radar, GPS, etc.), processors (e.g., CPUs, GPUs, ASICs, etc.), and attitude control (e.g., reaction wheels, flight control surfaces, etc.). Sensor data are processed in real time using computer vision (CV) and machine learning (ML) algorithms like convolutional neural networks (CNNs) [66, 67] to determine the ABU position relative to the target aircraft [59]. The ABU maintains communication with the launch or landing site and the target aircraft for safety.

ABU conceptual design parameters include the rotor diameter and mass; blade count and shape; electric motor characteristics; the ADACS characteristics (including sensor data, computer hardware, software, and control capability); and battery characteristics (e.g., volume). For the blended-wing ABU concept, additional design parameters include the wing shape, the wing area, and additional flight control surfaces. Selection of these design parameters depend on the category of aircraft supported (e.g., UAM, utility, or RAM aircraft). Overall design constraints inform these selections to minimize energy expended during travel between the host aircraft and a charging site. This work quantifies the benefits of ABUs in eVTOL UAM operations, motivating subsequent work exploring these design parameters.

B. ABU applications

This work evaluates four applications of ABUs supporting electric aircraft operations illustrated in Figure 5. These applications span the takeoff, cruise, and landing phases of flight. Each application either increases effective energy storage, reduces excess mass, improves safety, or provides some combination of each of these improvements.

Assisted takeoff. Takeoff represents one of the most power-intensive portions of a flight plan (see Figure 4a). After ABUs assist takeoff, they detach and return to the launch site to recharge. This application (i) avoids expending energy from permanent onboard energy buffers of the host aircraft, and (ii) reduces the mass of the aircraft by detaching depleted ABUs before cruise. The short lateral distance to the takeoff site supports key ABU design factors described in Section IV.A, and the ABUs may be charged independently from other aircraft — improving airport operations [68].

In-flight detachment. During cruise, energy stored in batteries depletes, but the mass of the batteries does not change. In-flight detachment of ABUs as they near depletion removes unused mass from the aircraft and improves flight efficiency. For a direct-rendezvous ABU design, detachment must occur near a potential landing site for the ABU. The ABU directly descends to this site using a small amount of charge. This application also supports airport operations and economics, allowing intermediate sites to participate in a flight that otherwise would have simply flown overhead.

In-flight rendezvous. Also during cruise, an ABU directly ascends from a site near the flight path to rendezvous with the aircraft. This ABU contains a nearly-charged battery which, once connected with the host aircraft, provides the equivalent of in-flight refueling. This application extends the range of an electrified aircraft, addressing the significant energy capacity challenge of battery-based flight. Economic and operational benefits arise from this application as well; battery charging distributes across a larger geographic area, and additional airports participate in supporting a flight.

Assisted landing. Aircraft should launch with all energy required to land, plus margin to address flight plan deviations or unexpected events. In rare cases, an aircraft experiences onboard failures eliminating margins and placing the aircraft at risk. ABUs improve safety and assurance in these situations. The target airport deploys ABUs to rendezvous with the troubled aircraft and replace the capability lost by the failure.

Supporting the most common UAM flight durations. When eVTOLs are sized, they target a maximum distance. However, most UAM flights span much shorter distances than this maximum. By shedding excess battery mass before a shorter flight, ABUs better support the common case of short-distance flights in UAM operations.

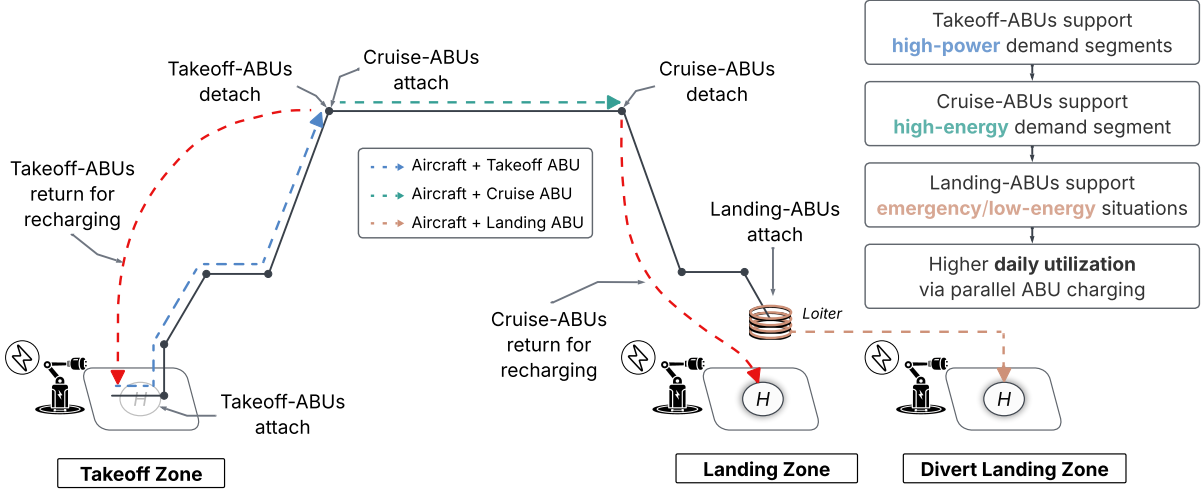


Fig. 5 This work evaluates (i) assisted-takeoff ABUs; (ii) assisted-cruise ABUs; and (iii) assisted-landing ABUs. By charging ABUs during eVTOL flight, these aircraft complete more flights per operating period.

V. Methodology

This work quantifies the benefits of ABUs for eVTOL UAM missions. To accomplish this aim, we use the *evtolpy* simulator (see Section III). This work also models queuing effects of multiple ABUs charging while eVTOLs are in flight. The simulator was applied to three eVTOLs: the Archer Midnight, the Joby S4, and the Supernal S-A2. These aircraft represent a mix of vectored-thrust and hybrid multi-rotor configurations, providing a useful cross-section of vehicle characteristics relevant to UAM services. For each aircraft, this work simulates missions at two cruise altitudes (1500 ft and 3000 ft) and three route lengths (30 mi, 45 mi, and 60 mi), resulting in 18 case studies per vehicle. Applying the ABU framework across this set allows the study to evaluate assisted-takeoff, assisted-cruise, and assisted-landing benefits over a broad range of vehicle types and operating conditions. Short-range missions of this scale align with proposed UAM operations, which typically involve urban and suburban routes.

A. Modeling eVTOL mission energy

A mission-level model computes eVTOL power and energy requirements across UAM mission segments illustrated in Figure 2. For each segment, *evtolpy* calculates instantaneous power demand. Hover and vertical-transition phases enforce a lift-equals-weight condition with induced power estimated from rotorcraft momentum theory. Wingborne flight segments apply a thrust-equals-drag balance using aerodynamic coefficients consistent with the aircraft design. These calculations utilize baseline aircraft parameters and output values used to size the ABUs for a given mission profile. Power demand scales with vehicle mass, ensuring that any mass changes due to ABU attachment or detachment propagate through the energy consumption model.

The mission model outputs a time-history of electric power, which is integrated over each segment to obtain segment-level energy. This process mirrors established design frameworks for electric aircraft while extending them to resolve eVTOL-specific performance drivers (e.g., rotor-induced hover power, high-power transition phases, and altitude-dependent cruise efficiency). This work evaluates three representative eVTOL aircraft — Archer Midnight, Joby S4, and Supernal S-A2 — at two cruise altitudes (1500 ft and 3000 ft) and three mission ranges (30 mi, 45 mi, 60 mi).

The *evtolpy* simulator models battery behavior using a constant-current, constant-voltage (CC-CV) charging framework informed by lithium-ion battery studies [69–73]; see equations below. During discharge, energy subtracts directly from available integrated battery capacity while *evtolpy* tracks the depth of discharge (DoD). During charging, a CC phase applies until 80% state of charge (SOC), followed by a CV hold until the termination current ratio reaches 5%. All batteries, including ABUs, utilize a 115-kW AC charger operating at 95% efficiency. This uniform charging model allows for consistent comparison of charge times across baseline eVTOL missions and ABU-supported missions.

$$t_{CC} = \frac{(SOC_{CC} - SOC_{in})}{100} \frac{C}{I_0} \quad t_{CV} = \frac{(SOC_{fin} - SOC_{CC})}{100} \frac{C}{I_0} \frac{\ln(1/k)}{1-k} \quad t_{tot} = t_{CC} + t_{CV}$$

B. Modeling assisted-takeoff ABUs

Assisted-takeoff ABUs act as supplementary power sources that attach to an eVTOL before takeoff and detach before cruise. These ABUs mitigate the power demand of hover climb, transition climb, and departure procedures. As a result, the eVTOL may be designed with a smaller main battery. Each assisted-takeoff ABU provides 15 kWh of usable mission energy, which covers all pre-cruise segments with margin and reserves 6 kWh for post-detachment operations. This reserve includes energy for vertical descent and horizontal cruise to the departure site with margin.

Upon detachment at cruise altitude, the ABU conducts its return flight, modeled as horizontal cruise at 30 m/s followed by vertical descent at 5.1 m/s. After landing, the ABU recharges under the CC–CV model and returns to the pool of takeoff ABUs. By offloading early-mission energy, the takeoff ABU reduces main-battery depletion and shortens main-pack recharge time, thereby improving achievable flight cadence.

C. Modeling assisted-cruise ABUs

Cruise ABUs detach during flight and, while attached, reduce the main battery depth of discharge (DOD) by supplying a portion of the cruise-segment energy. Each ABU attaches at the beginning of cruise and supplies power until either its internal cruise-energy capacity depletes or the baseline cruise segment ends. In this work, we sweep usable mission energy of cruise ABUs from 5 kWh to 50 kWh.

To ensure adequate performance during rendezvous and return, each cruise ABU reserves 12 kWh for self-operation, about 2× that of takeoff and landing ABUs, reflecting its need to ascend for rendezvous and descend to a different landing site after detachment. The `evtolpy` framework determines ABU mass using battery specific energy with structural and integration overheads of 20% and 5%. This mass increases aircraft TOGW and impacts cruise-segment power needs.

After detaching at cruise altitude, the ABU performs a return trajectory analogous to the takeoff ABU but directed toward a destination charging site. Once recharged, the ABU is available for future missions. This approach enables cruise ABUs to be evaluated not only for energy benefits but also for operational impact on multi-flight simulations.

D. Modeling assisted-landing ABUs for divert scenarios

Assisted-landing ABUs contribute to flight safety during low-energy arrivals or unexpected landing-zone (LZ) restrictions. Unlike takeoff and cruise ABUs, which enhance mission efficiency, landing ABUs enhance flight assurance. An assisted-landing ABU attaches after the descent segment in response to an aircraft unable to ensure a safe landing.

To evaluate feasibility across aircraft types and altitudes, we standardize a divert profile. This profile consists of a 60 s hover hold, a 6 mi divert cruise at 48.7 m/s, and a hover-descent segment lasting 20 s to 40 s depending on cruise altitude. The ABU must provide sufficient usable mission energy to complete all three divert segments. As with other ABU types, the landing ABU reserves 6 kWh for self-operation, and its usable mission energy is swept from 5 kWh to 50 kWh to determine minimum energy capacity for safe divert-and-land.

E. Daily operations and ABU queuing simulation

All battery charging, including that of the main aircraft battery and all ABUs, is modeled with the same CC–CV model. The integrated battery of the eVTOL consists of 800 V nominal voltage. ABUs employ a 400 V nominal voltage, representing a modular mid-voltage class compatible with lightweight detachable connectors, 400 V power electronics, and safe human-machine interactions during autonomous rendezvous.

The `evtolpy` simulator models turnaround time using a two-state mixture distribution presented by Bruehl [74], in which non-charging ground operations require either 13 min (no maintenance) or 21 min (light maintenance). By adopting a maintenance probability of 0.5, the expected turnaround time becomes 17 min with a standard deviation of 4 min. This representation allows the operational model to reflect the variability of UAM ground processes.

To evaluate mission-level energy benefits in an operational context, this work models an 8 h operations timeline. During this time, the aircraft repeats the same mission subject to constraints on ABU availability and main-battery recharge time. The simulation maintains separate timelines for each ABU and the eVTOL. Each ABU cycles through attachment, power-delivery, detachment, return flight, recharge, and idle phases. The aircraft may depart only when its main battery has fully recharged and when both a takeoff ABU and a cruise ABU are fully charged and available at the landing zone. Any delay caused by ABU unavailability is recorded as an ABU-driven bottleneck. Because ABUs require finite return-and-recharge time, the simulation captures queuing dynamics and resource contention effects. These dynamics determine the achievable aircraft flight cadence and the number of ABUs required per LZ to avoid bottlenecks.

This methodology integrates per-segment eVTOL energy modeling, detachable ABU operations, battery-charging

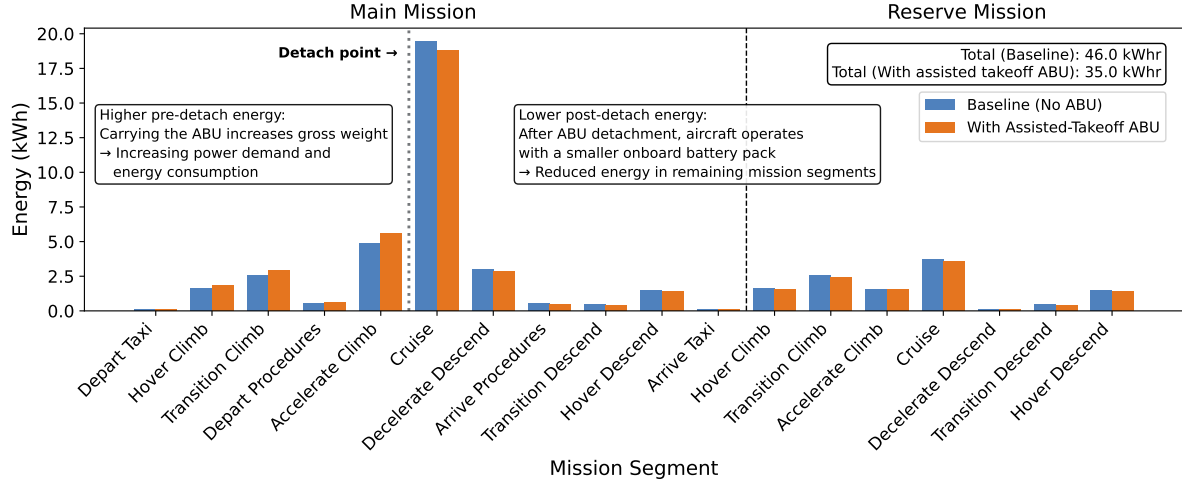


Fig. 6 The segment-level energy profile of Joby S4 with and without an assisted-takeoff ABU for a 30 mi mission at 3000 ft. An assisted-takeoff ABU reduces mission energy by 24% by enabling a smaller onboard battery.

models, and queuing dynamics into the `evtolpy` simulation framework. This simulator supports evaluation of ABUs for their impact on per-mission energy consumption, mission safety, and flight throughput. This comprehensive framework provides a basis for quantifying the performance and economic impact of ABU-enabled eVTOL operations.

VI. Evaluation and Results

A. Assisted-takeoff ABUs reduce mission energy requirements

Figure 6 shows the impact of assisted-takeoff ABUs on the mission segment energy profile of the Joby S4 during a 30 mi mission at 3000 ft. As expected, the eVTOL exhibits slightly higher energy consumption during pre-cruise segments due to the added mass of the ABU. Once the ABU separates at the start of cruise, the aircraft proceeds with a smaller onboard battery and reduced gross weight, thereby lowering the energy requirements of subsequent segments. The assisted-takeoff ABU enables an overall reduction in total mission energy of 24% (from 46 kWh to 35 kWh).

A further insight from this segment-level breakdown is that assisted-takeoff changes not only how much energy the aircraft uses, but also when the energy is drawn from the main battery. By shifting a substantial portion of the early mission demand away from the integrated battery, the aircraft begins its cruise phase with less heating and electrochemical stress on the integrated battery. Because lithium-ion efficiency and internal resistance vary with both SOC and thermal conditions, entering cruise in a more favorable operating state can yield additional efficiency gains not captured solely by mass differences. This fact suggests that assisted-takeoff could benefit battery longevity and thermal management in addition to reducing total mission energy. These secondary effects become increasingly important for high-frequency UAM operations in which batteries are cycled repeatedly over short intervals.

Figure 7 quantifies these effects across all simulated eVTOLs, mission distances, and cruise altitudes. Assisted-takeoff ABUs reduce total mission energy by 7 – 24%. The largest reductions appear for shorter-range missions. This trend arises from the fixed nature of takeoff and climb energy demand: because these segments require the same energy regardless of mission length, offloading pre-cruise energy has a greater relative impact when the cruise segment is short. For longer missions, cruise energy dominates, reducing the proportional impact of assisted takeoff even as the absolute energy savings remain significant. This pattern persists across all three eVTOL configurations despite different rotor architectures and hover efficiencies. In each case, reducing early-segment battery load and allowing the vehicle to complete most of the mission with a lighter onboard battery yields significant reduction in mission-level energy.

Even modest reductions in integrated battery mass are meaningful for eVTOL sizing and performance, because battery mass contributes significantly to gross weight and strongly influences hover power, structural load, and charge time. The Figure 7 results highlight a key benefit of assisted-takeoff ABUs: decreasing integrated battery size relaxes sizing constraints, enabling lighter, more operationally-efficient vehicle configurations without altering the mission.

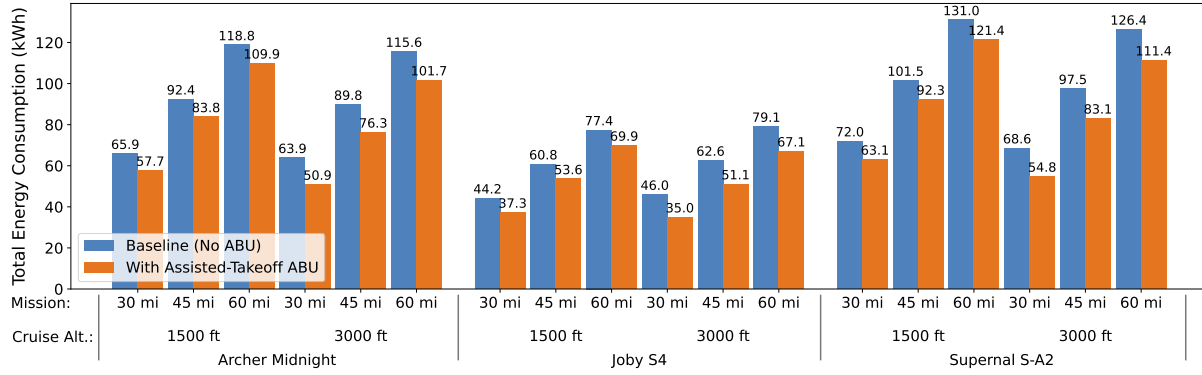


Fig. 7 Per-mission energy consumption with and without assisted-takeoff ABUs. Across all aircraft, mission ranges, and cruise altitudes, assisted-takeoff ABUs reduce total mission energy consumption by 7–24%.

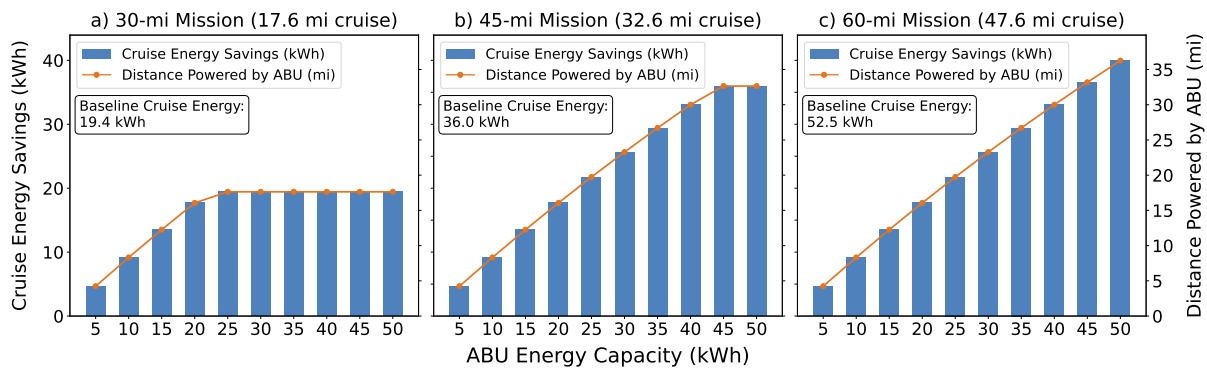


Fig. 8 Increasing assisted-cruise ABU energy capacity supports more cruise-segment energy savings and longer ABU-powered distances for Joby S4 at 3000 ft. Benefits saturate once the ABU powers the entire cruise segment.

B. Assisted-cruise ABUs reduce integrated battery depth-of-discharge

Figure 8 illustrates how assisted-cruise ABUs affect cruise-segment energy for Joby S4 missions of 30 mi, 45 mi, and 60 mi at 3000 ft. Increasing ABU energy capacity yields larger cruise-segment energy savings and longer distances powered by the ABU. The results show a clear saturation point: once the usable energy of the ABU is sufficient to cover the cruise segment, additional ABU capacity no longer increases the energy savings or distance flown under ABU power. The 30 mi mission reaches this point at 25 kWh, while the mid-range mission requires 45 kWh and the long-range mission requires more than 50 kWh. These results show that ABU sizing couples tightly to mission length.

Figure 8 also shows that the shape of each curve is governed by the underlying mission cruise duration. The 30 mi case reaches saturation quickly, because only 17.6 mi of cruise must be powered before the ABU fully replaces the integrated battery, whereas the 45 mi and 60 mi missions contain 32.6 mi and 47.6 mi of cruise, respectively, and thus require more ABU energy to achieve the same effect. This fact shows that ABU benefits scale with cruise duration rather than total mission distance, and it explains why missions with longer cruise segments exhibit nearly linear gains well beyond the point at which the short-range missions plateau. The comparison also shows that, even when an ABU cannot power the entire cruise, partial coverage still displaces a significant portion of the baseline cruise draw, indicating that the operational value of an assisted-cruise ABU is not limited to complete segment replacement.

Figure 9 extends the analysis to all flight profiles. Assisted-cruise ABUs reduce integrated battery depth-of-discharge, allowing short-range missions to be powered almost entirely by the ABU. For mid-range missions, the ABU powers a significant portion of cruise. ABU benefits are most muted for long-range missions. However, the significant reduction in DOD shows that assisted-cruise ABUs remain valuable even when they partially power the cruise segment.

Table 3 summarizes this effect by reporting integrated battery LZ DOD. Across all aircraft, altitudes, and ranges, assisted-cruise ABUs significantly reduce DOD. In 30 mi missions, DOD falls from a baseline of 75% – 79% to just 16% – 19%. For 45 mi and 60 mi missions, the ABU no longer fully covers cruise, yet DOD still drops to just

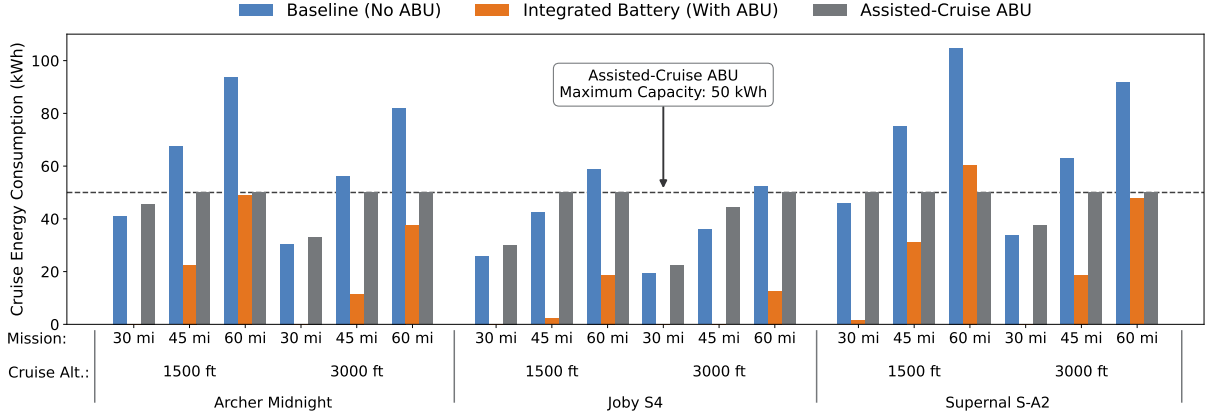


Fig. 9 Cruise energy with and without ABUs, which fully power short missions and reduce longer mission DOD.

24% – 41%. These improvements remain substantial when compared to the baseline DOD of 85% – 88%. These results indicate that assisted-cruise ABUs provide benefits across a wide range of missions. They reduce integrated battery DOD and allow short-range flights to complete with minimal use of the integrated battery. These improvements enhance operational efficiency, support smaller battery sizes and alleviate degradation, all while increasing UAM flight cadence.

C. Assisted-landing ABUs enable unexpected divert missions

Figure 10 evaluates the performance of assisted-landing ABUs in divert scenarios, i.e., when the eVTOL performs a hover hold, executes a divert cruise, and completes a hover-descent at an alternate LZ. The left panel shows results for the Supernal S-A2 at 1500 ft. ABUs with insufficient usable mission energy cannot support a divert, which is reflected by zero loiter time after the mandatory phases. Once ABU capacity reaches the feasibility threshold, the eVTOL completes the divert mission and supports additional loiter time. This surplus grows with more ABU capacity, rising from six seconds at the minimum feasible size to nearly 70 s at the largest size considered.

The right panel shows these findings across all aircraft and cruise altitudes. The required ABU size depends on vehicle configuration and operating altitude, because both factors influence hover power, vertical descent energy, and the aerodynamic efficiency of the divert cruise. For example, Archer Midnight requires approximately 40 kWh at 1500 ft and 45 kWh at 3000 ft to complete the divert profile, while Joby S4 requires 30 kWh to 35 kWh. Supernal S-A2 shows similar trends with required capacities from 35 kWh to 40 kWh. These results confirm that divert mission energy needs are substantial, driven by the hover-hold segment and the vertical landing segment, both of which scale with gross weight and altitude. Assisted-landing ABUs must be sized for emergency procedures rather than mission efficiency.

These results show that assisted-landing ABUs add resilience to eVTOL operations by providing divert capability under low SoC. The ABU ensures that aircraft retain sufficient energy to perform a complete divert flight, and larger ABU sizes introduce additional loiter margin essential to real-world uncertainties. Assisted-landing ABUs serve as a

Table 3 Onboard battery DOD across eVTOLs, altitudes, and ranges for baseline and assisted-cruise ABU flights. For Archer Midnight and Supernal S-A2, only the 30 mi mission can be fully-powered by the largest ABU.

		LZ Depth of Discharge (%)					
Aircraft	Altitude (ft)	Baseline (No ABU)			With Cruise-ABU		
		30 mi	45 mi	60 mi	30 mi	45 mi	60 mi
Archer Midnight	1500	78%	84%	88%	16%	36%	50%
Archer Midnight	3000	75%	82%	86%	28%	33%	48%
Joby S4	1500	77%	84%	87%	19%	17%	35%
Joby S4	3000	75%	82%	85%	33%	24%	35%
Supernal S-A2	1500	79%	85%	88%	17%	41%	55%
Supernal S-A2	3000	77%	84%	88%	28%	39%	53%

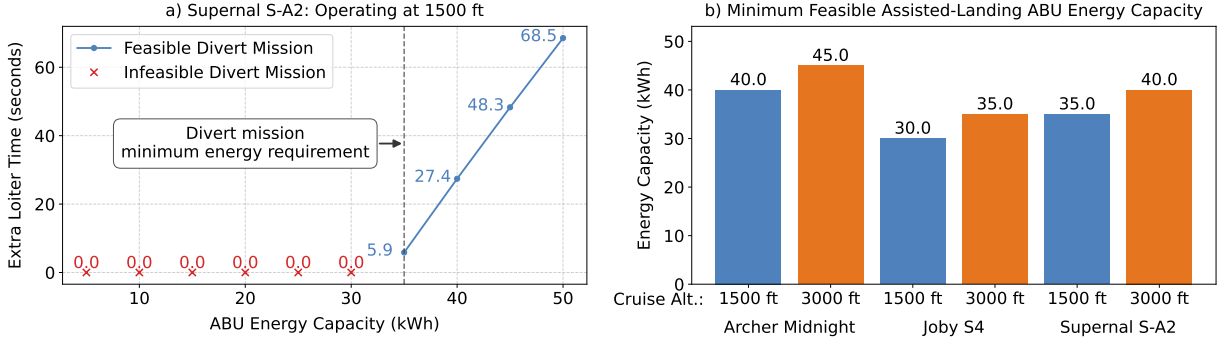


Fig. 10 Assisted-landing ABU benefits for divert flights. Left: Larger-capacity ABUs enable more loiter time. Feasible cases appear only when ABUs contains sufficient energy for the divert-cruise, hover hold, and hover-descent segments. Right: Minimum ABU energy for each eVTOL and altitude to make a divert feasible.

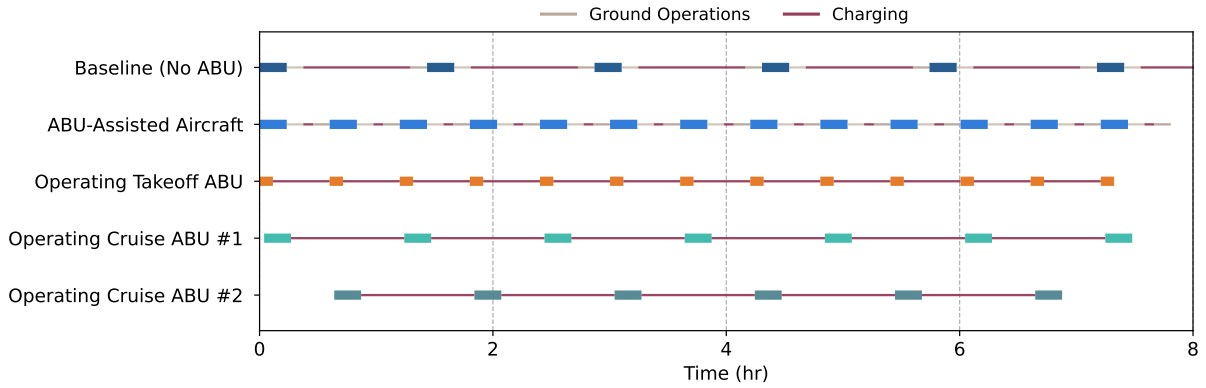


Fig. 11 An operating timeline for baseline and ABU-supported missions over an 8 hr window for Archer Midnight (30 mi mission at 1500 ft). Assisted-takeoff ABUs shorten aircraft charge cycles, while two cruise ABUs operate in rotation to avoid charging bottlenecks. ABUs enable a higher flight cadence than the baseline with staggered ABU charging ensuring continuous mission support throughout the operating period.

safety-enhancing complement to the performance-oriented benefits of assisted-takeoff and assisted-cruise ABUs. They provide a protective buffer that increases operational robustness in scenarios such as Air Traffic Control (ATC)-induced holding, weather-related delays, or unexpected congestion at the primary LZ.

D. ABUs increase eVTOL flight cadence

The operating sequences shown in Figure 11 reveal how ABUs improve the temporal rhythm of flight, charging, and ground operations during an eight-hour service window. In the baseline case, Archer Midnight cycles between flight, ground activities, and long battery charging periods — leaving a significant fraction of this time window idle. An assisted-takeoff ABU shortens charging intervals, because the aircraft arrives with a higher SoC. With the addition of two cruise ABUs, flight cadence increases dramatically. The cruise units rotate through attach, detach, return, and charging cycles in a staggered pattern, ensuring that a fully-charged unit is always available. This rotation eliminates the charging bottleneck when a single cruise ABU must wait for charging to complete. The resulting timeline exhibits a more continuous and tightly-packed mission cadence with fewer gaps between departures and fewer instances of the eVTOL waiting for energy to become available.

Beyond increasing cadence, ABUs help to resolve a logistics bottleneck limiting eVTOL fleets. In the baseline configuration, aircraft turnaround time is dictated almost entirely by integrated battery charge time; even small variations in DOD or charging rate result in missed departures. With ABUs, charging shifts from a serial process to a parallelized system in which multiple devices charge simultaneously. This parallelism makes the overall operation more tolerant of delays in any one element, whether caused by temperature-dependent charging limits, temporary charger unavailability,

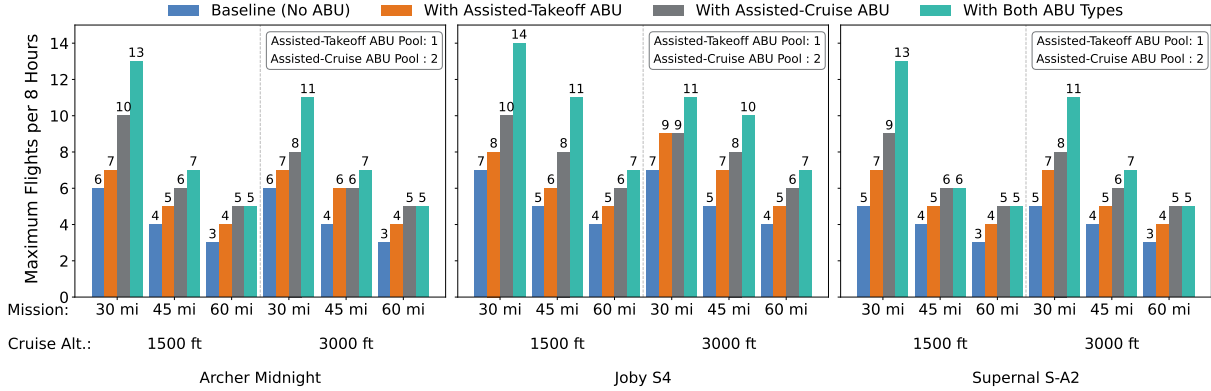


Fig. 12 Effects of assisted-takeoff, assisted-cruise, and combined ABU usage on eVTOL flight cadence over an 8 hr period. Deploying two cruise ABUs removes the charging bottleneck, enabling more frequent turnarounds. Assisted-takeoff ABUs modestly increase flight count by shortening charge time, cruise ABUs provide additional gains by reducing main-pack depletion, and the combined configuration yields the largest improvements relative to the baseline — particularly for shorter missions and lower-altitude operations common to UAM.

or deviations in ground-operation times. The timeline in Figure 11 illustrates this observation: departures are governed not by a single battery charge, but by the availability charged ABUs. This redistribution of operational risk away from the integrated battery system is particularly valuable in high-utilization UAM scenarios, where consistent cadence is often more important than maximizing the performance of a single flight.

Figure 12 summarizes the effect on mission throughput by reporting the maximum number of flights within eight hours across all aircraft, mission lengths, and cruise altitudes. Assisted takeoff alone increases throughput by reducing main battery charge time, although the gains are modest. Cruise ABUs increase flight count more significantly, because cruise energy dominates. Lowering DOD accelerates charging. The most substantial improvements occur when applying both ABU types. Assisted-takeoff ABUs alleviate early-flight energy consumption, while assisted-cruise ABUs charge asynchronously. This configuration produces the largest flight counts, particularly for short missions common in UAM.

These results show that ABUs improve operational performance by decreasing mission energy and restructuring how energy is supplied and replenished over time. Assisted-takeoff ABUs reduce the initial draw on the main battery, assisted-cruise ABUs alleviate the dominant mid-mission load, and the presence of multiple ABUs prevents queueing delays that would otherwise limit departures. The combined effect is a more resilient and higher-throughput operational profile, enabling eVTOL aircraft to sustain flight cadences well beyond what integrated battery systems support alone.

VII. Conclusion

This work presents autonomous battery units (ABUs) as a practical improvement to eVTOL architectures by enhancing UAM mission operational performance. By storing portions of mission energy in detachable units that rendezvous, attach, detach, and charge independently, ABUs mitigate coupling between onboard battery mass, mission range, and charge time. This work evaluates three ABU types. Assisted-takeoff ABUs alleviate the high power of early mission segments, enabling a smaller onboard battery mass. Assisted-cruise ABUs supply power during the most energy-intensive portion of the flight, reducing depth of discharge. Assisted-landing ABUs provide a dedicated reserve for divert scenarios, offering a safety margin under low-SOC conditions. These capabilities reduce mission-level energy requirements, shorten charge time, and eliminate charging bottlenecks when multiple ABUs operate in rotation. As a result, eVTOLs to achieve higher flight cadence over typical operating times.

These results show that ABUs offer a near-term pathway to improving the operational tempo of eVTOLs. Future work includes ABU structural and aerodynamic design, battery performance and reliability under repeated flights, autonomous docking and navigation, and the impact of ABU mass on aircraft structure and sizing. We aim for this work to inform regulatory and certification pathways for detachable energy systems. As UAM operations mature, these results suggest that ABUs contribute to improved practicality, robustness, and economic viability of eVTOL UAM services.

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